OSMonto

An Ontology for OpenStreetMap tags

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Outline

- Motivation
- Related Work
- Ontologies
- OSMonto
- Applications: DO-ROAM
- Conclusions

Motivation

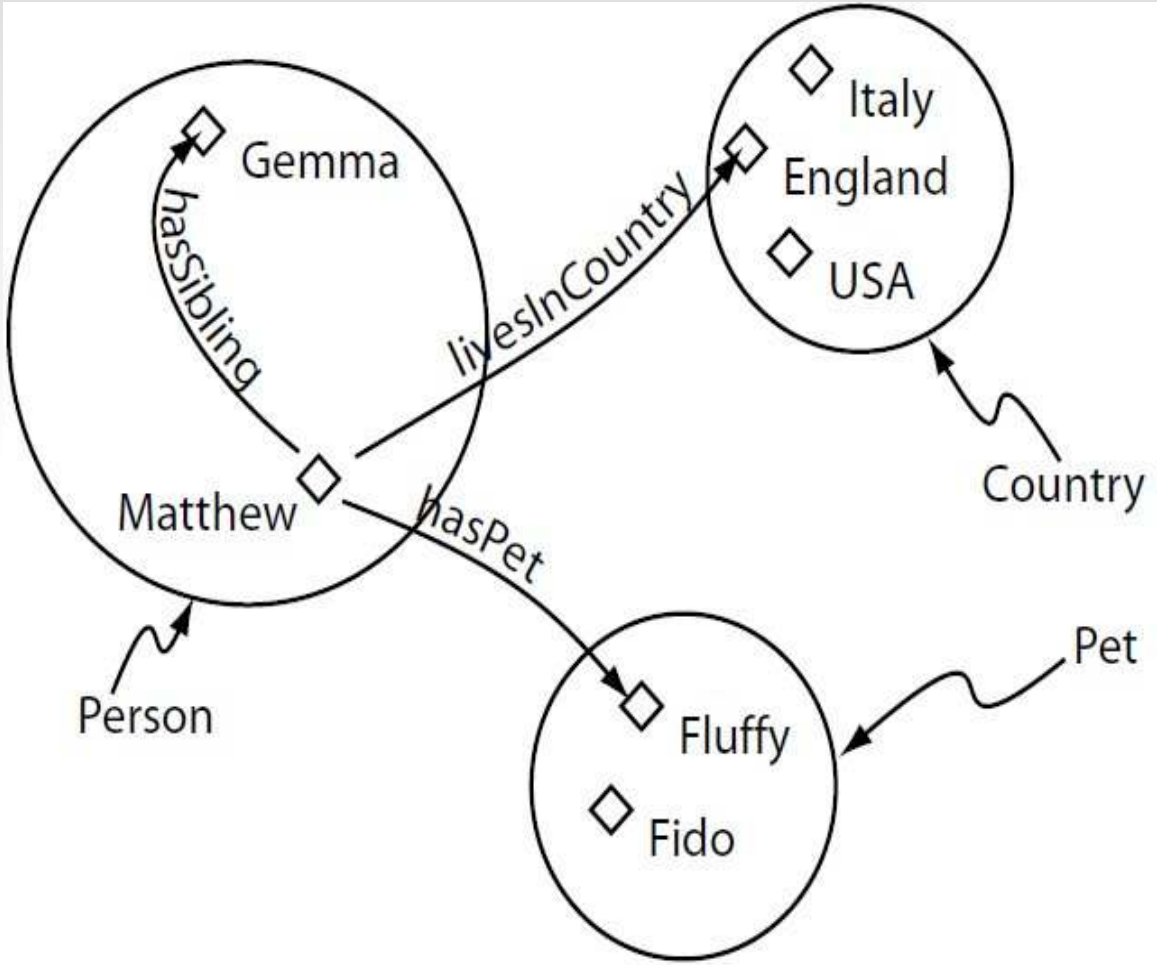
- Hierarchical representation of OSM tags
- Provides a better overview (via tools like Protégé) and aims towards automatic maintenance
- Ontological semantics via mappings: different versions or semantically related tags can be united
- Ease access to the tags for search and navigation in an OSM map (DO-ROAM)

Related Work

- Taginfo *(http://taginfo.openstreetmap.org/)*
- Tagwatch *(http://tagwatch.stoecker.eu/)*
- OSM wiki *(http://wiki.openstreetmap.org/wiki/Map_features)*

Ontologies

- Capture knowledge about some domain by describing its concepts and the relationships between them
- Wide range of applications in computer science

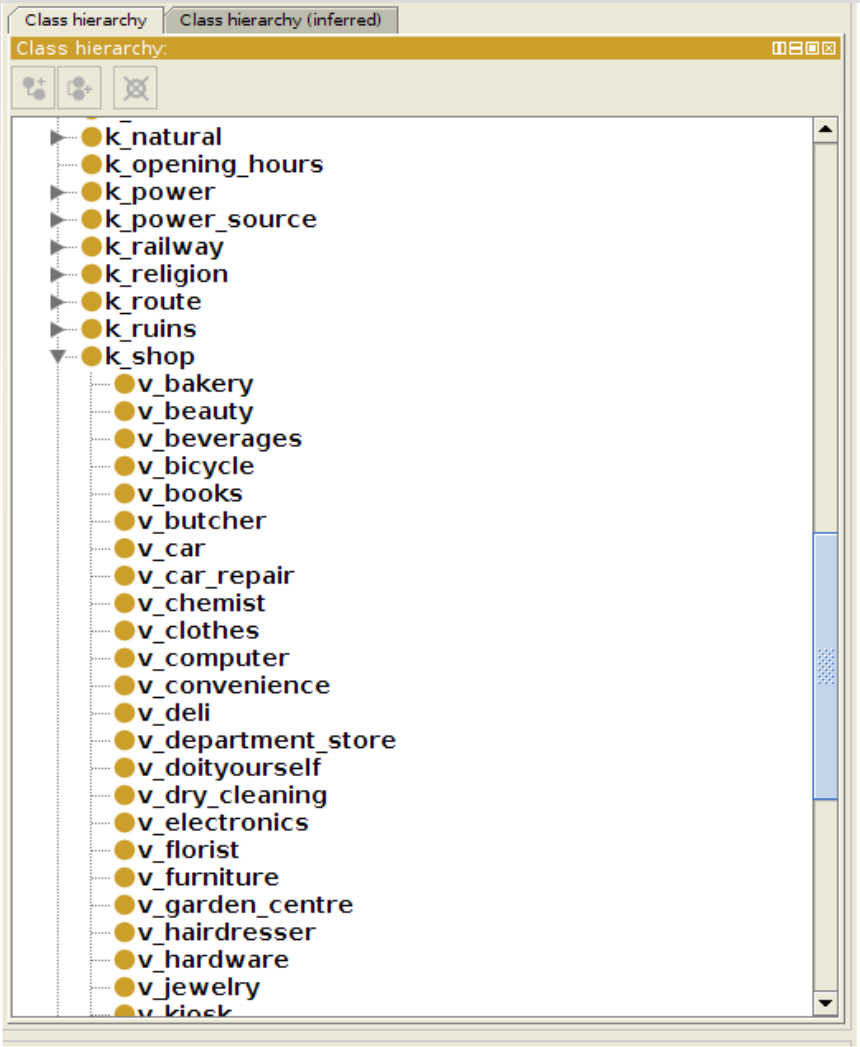


OSMonto

- OSM tags are of form

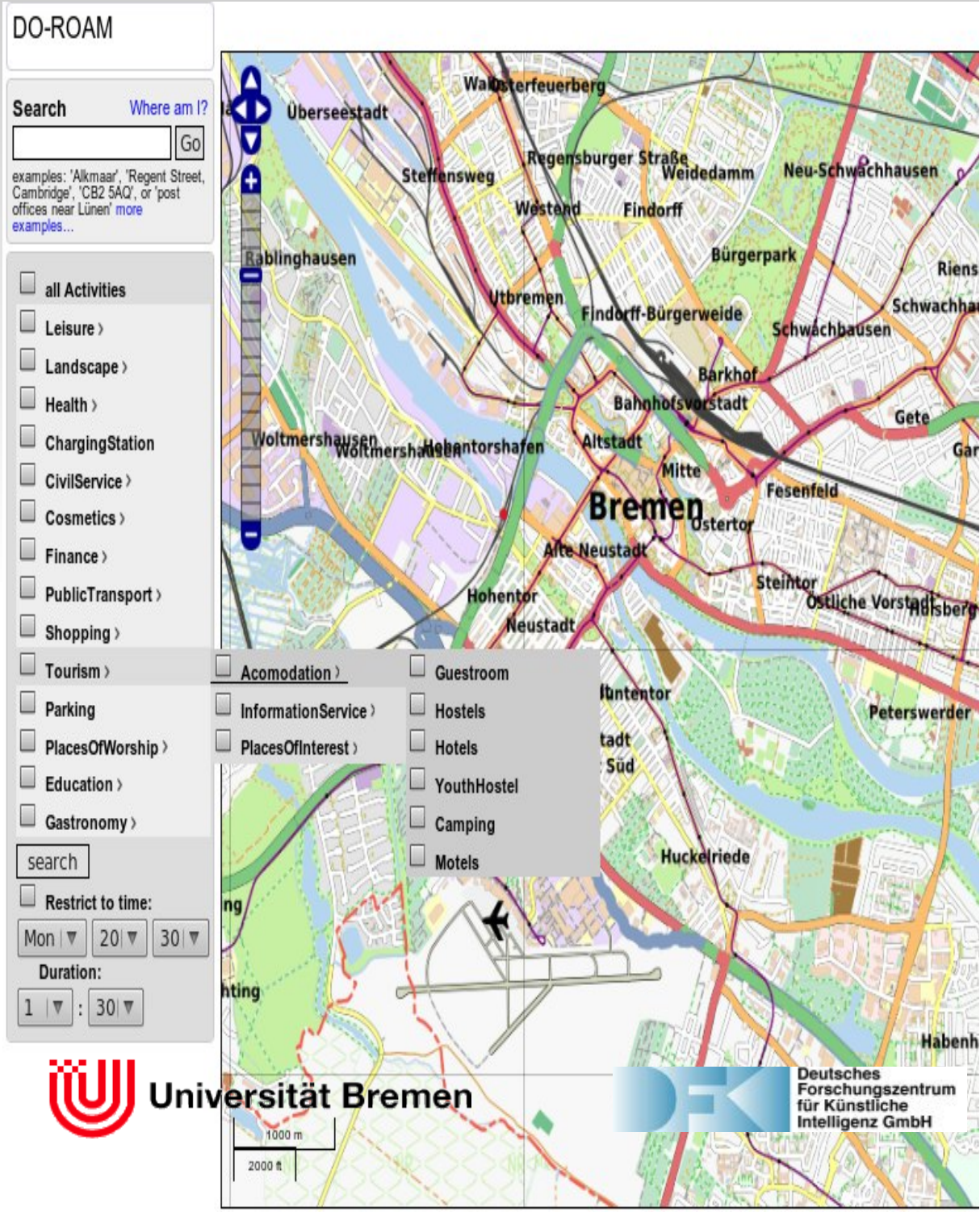
  < tag k = “key” v = ”value”>

- Keys become superconcepts of their values
- Some tags can be used only in presence of other tags => roles
- Dynamic aspect: OSM community can introduce new tags or modify existing ones
- Threshold for selecting a tag as relevant => errors are reduced
- http://wiki.openstreetmap.org/wiki/OSMonto

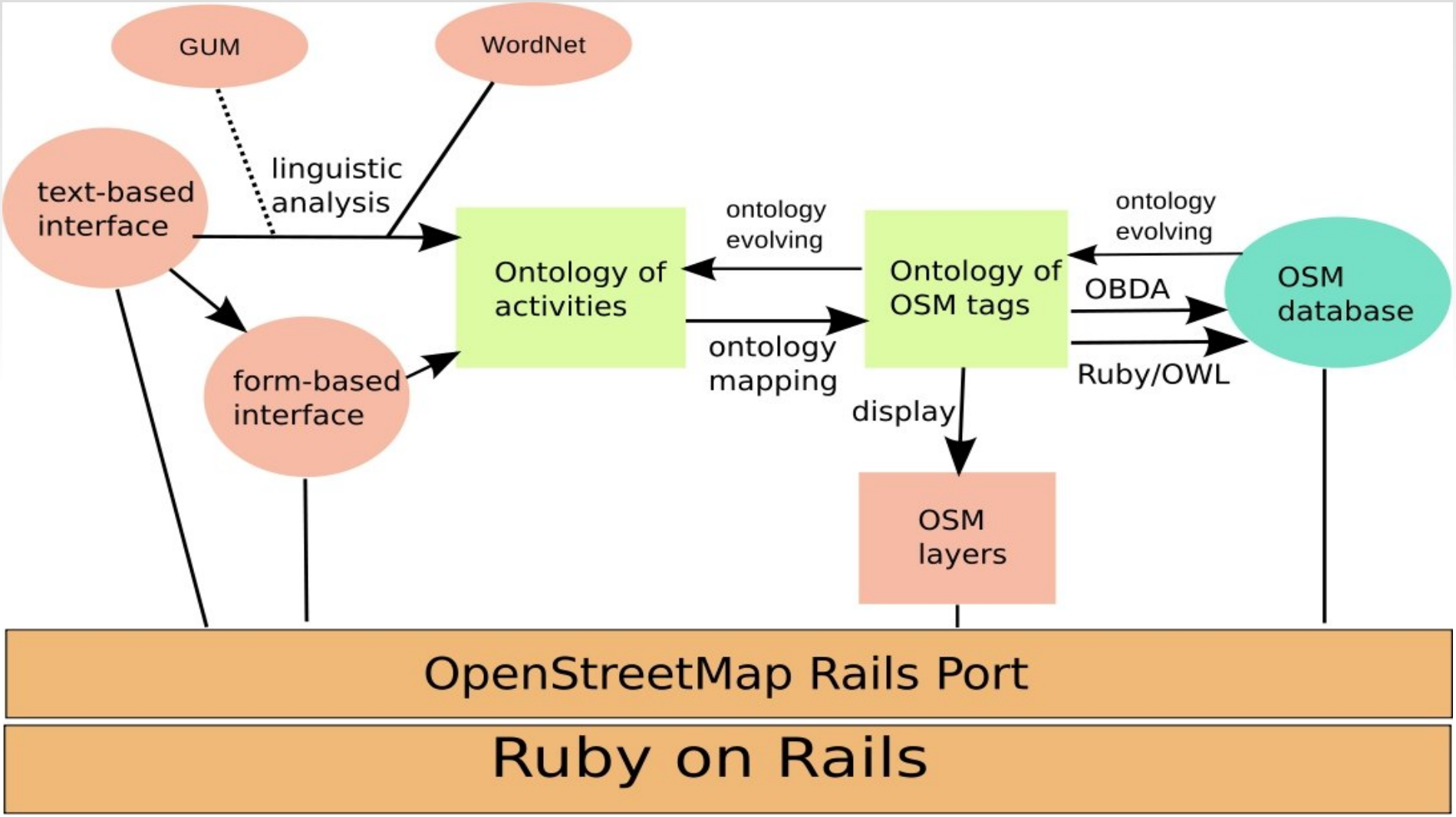


DO-ROAM

- focus on **usability**
- **evolving** set of activities organized as an ontology related to OSMonto
- **linguistic analysis** (synonyms for now) for free text search
- **OpenStreetMap** data: open source, constantly developing, sometimes better than Google Maps; partly language-agnostic



General Tool Architecture



Route Planning System



Conclusions and future work

- OSMonto provides structure for OSM tags and means of adding additional semantics to them
- Reference for the OSM community and possible insights on how to extend and structure OSM tags
- Extend ontology with more roles and concepts to allow to infer missing information using OBDA (restaurant with chinese name or in China Town -> Chinese cuisine)